

All-Empowering Search

We live and breathe search—the Internet would be pretty much useless without it. Whither headed?

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“**O**ur two primal urges—to know the world and have the world know us—will flower as never before.” That’s Quentin Hardy, Silicon Valley bureau chief of *Forbes* magazine. As a prediction for the coming state of the Web, we must agree.

Information is increasing at a pace we as a race just cannot handle. The search engines are here to help. They will be there, in better avatars, as the nature of content changes—witness how non-textual information is burgeoning. Think of image search, audio search, video search: our current text-box-entry interface is quite like the DOS prompt. So visualise, with all the imagination you can muster, what would happen if a Windows or Mac-like evolution were to happen—hum a tune and get your song. Upload an image and get similar ones, Riya-style. (www.riya.com; “Find an item you like and **Like.com** will show you results that are visually similar.”) Speak out a video description and get your results...

Picking up on that last, few know about Google Voice Search—it’s been here for a year or so! You dial a number, speak out your search terms, wait a while, click a link, and there you go—your search results. Since there’s only one number to call, the system’s capacity is very limited—but this underlines the idea that in the future, we won’t necessarily be sitting at our Desktops and typing out queries. On the train, someone says, “So who won that match?” Out comes the handset, you say out “Australia England match,” and—the default might be “I’m feeling lucky”—you just see the headline you need to.

Jeremy Zawodny, “Technical Yahoo!”, said a while ago about the future of search: “Our local search is really just the beginning. (We need to) discern intent. When you search for a particular set of keywords, what’s your goal? Are you doing research? Looking to buy something? Trying to find a long lost friend? As search engines begin to understand our goals and motivations, they’ll be able to custom tailor the results...”

Yahoo!’s mysearch.yahoo.com is something like Google Personalized Search (more on that later), and the select-countries-only local.yahoo.com and its variants are indeed beginnings in the direction of more relevant results. As for intent, it gets closer to the core of the issue, and is quite an Artificial Intelligence (AI) problem; a stop-gap solution could be to explicitly ask the searcher, “What are you searching for?” followed by a list of possibilities as above. Well, not “long-lost friend,” but something like “people,” “places,” “research,” “Web entertainment,” and so on.

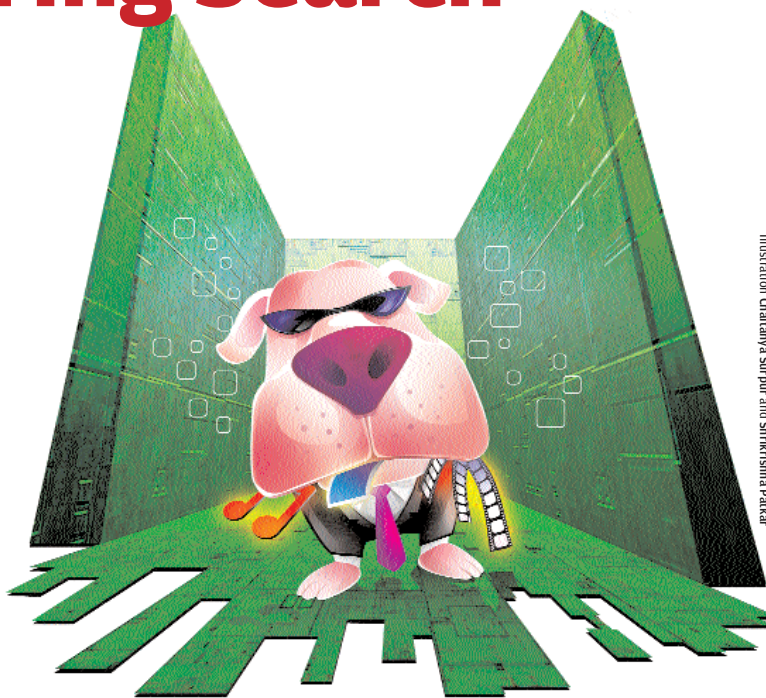


Illustration: Chaitanya Surpur and Shrikishna Patkar

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Disambiguation is as close to “intent” as we’ve gotten. We recall that about a year ago, Google began its—what’s it called now—“Refine results for:” thing. So you type in a drug name, and you get a list of more specific searches to choose from: “Drug uses,” “Side effects,” and many more. The kind of searches this is done for is only a few—like for drugs, like for city names. But we see this number increasing. Think of Wikipedia: when you type in a search word that is used in two or more contexts—like “Opera”—you come to a disambiguation page. How long before MSN, Yahoo!, and Google start off with their versions of context disambiguation?

As far as the raw algorithms go, they’re getting better in small, important steps. Note the “NEAR” operator we used in AltaVista: “NEAR” is the default in Google, as you’ve doubtless noticed. But the urgent need is better Machine Learning, better AI. We don’t see that happening anytime in the near future. But a lot can be done with what’s already been developed: Riya (though still abysmal in terms of volume of results) doesn’t depend on tags. That’s one of the mantras for search of the future: *don’t depend on tags*. Auto-tag instead. Use pattern recognition, face recognition.

From AI to Natural Intelligence: Yahoo! bought up <http://delicio.us/> in part to include bookmarked sites as a ranking criterion in search results. And then, what about Digg? Will more Digg mean higher rankings? In short, search engines actively using already-done *human activity* could very well happen.

Talking about human activity, people just don’t tag things: how many times have you provided a proper summary for your just-done Word document? How long ago did you give a picture